possible to replace brute force search with something more efficient?

If you think about it, for each point we are looking for its nearest neighbor among the set of centroids… and we already know a data structure or two to speed up this search!

But rather than Ss$^+$-trees, in this case the context would suggest we could try k-d trees, for three reasons: before reading them below, try to pause for a minute and think about why we could prefer k-d trees; if you can't find all three reasons, or if you don't have a completely clear idea about why these reasons hold, you can take a look at sections 9.4 and 10.1 that explains these concepts in more detail.

Anyway, the reasons why k-d trees would be better suited than Ss$^+$-trees for the nearest centroid search are:

- The size of the dataset (the centroids) is small and hence with very large probability it will fit into memory: k-d trees are the best choice in this case if…
- … the dimension of the search space goes from low to medium, and we know this is the case (if we have done our homework!) because k-means also suffer from the curse of dimensionality, and shouldn't be applied to high-dimensional datasets anyway;
- K-d trees can offer a theoretical worst-case upper bound that is better than brute-force search: `O(2d + d*log(k))` for `k` `d`-dimensional centroids; Ss$^+$-trees,